

L7: Account Management & Access Control

Account Management & Access Control

- Typically, MySQL creates the **root** user account during installation.
- This root user account has **full privileges** over the MySQL database server, giving it complete control over all databases, tables, users, and more.
- It's a good practice to use the root account for administrative functions exclusively.
- For other tasks, create a new user account and grant the necessary privileges.

CREATE USER statement

Syntax

```
CREATE USER [IF NOT EXISTS] <username>@<server>  
IDENTIFIED BY <password>;
```

Example

```
CREATE USER IF NOT EXISTS 'byte1'@'localhost'  
IDENTIFIED BY 'byte1';
```

```
Query OK, 0 rows affected (0.01 sec)
```

Note. From this slide, we will call <username>@<server> as `account_name`.

Show the current USER

Syntax

```
SELECT user  
FROM mysql.user;
```

Example

```
SELECT user FROM mysql.user;
```

user
byte1
mysql.infoschema
mysql.session
mysql.sys
root

```
5 rows in set (0.00 sec)
```

Access the created USER

CMD

```
mysql -u <username> -p  
Enter password: <password>;
```

For the root user, username = 'root'.

Note. To exit press 'ctrl + z'

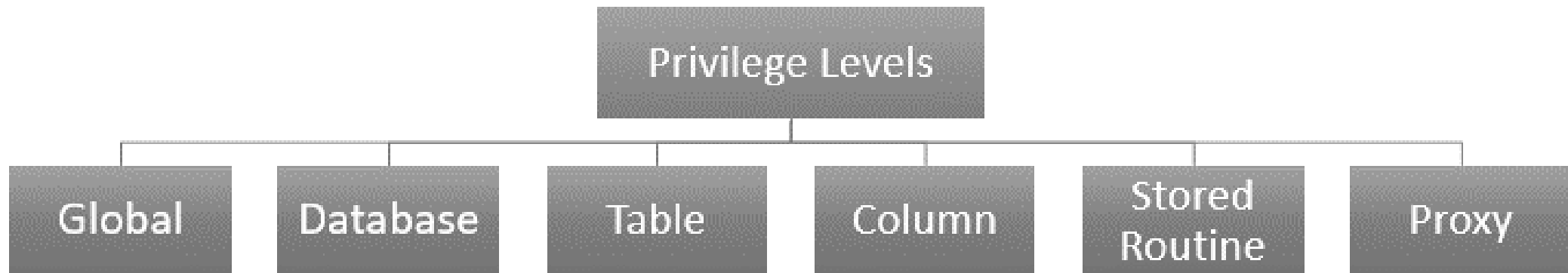
Example

```
C:\Users\MUICT>mysql -u byte1 -p  
Enter password: *****
```

```
Welcome to the MySQL monitor.  Commands end with ; or  
\g.  
Your MySQL connection id is 26  
Server version: 8.4.3 MySQL Community Server - GPL  
  
Copyright (c) 2000, 2024, Oracle and/or its affiliates.  
  
Oracle is a registered trademark of Oracle Corporation  
and/or its  
affiliates. Other names may be trademarks of their  
respective  
owners.  
  
Type 'help;' or '\h' for help. Type '\c' to clear the  
current input statement.  
  
mysql>
```

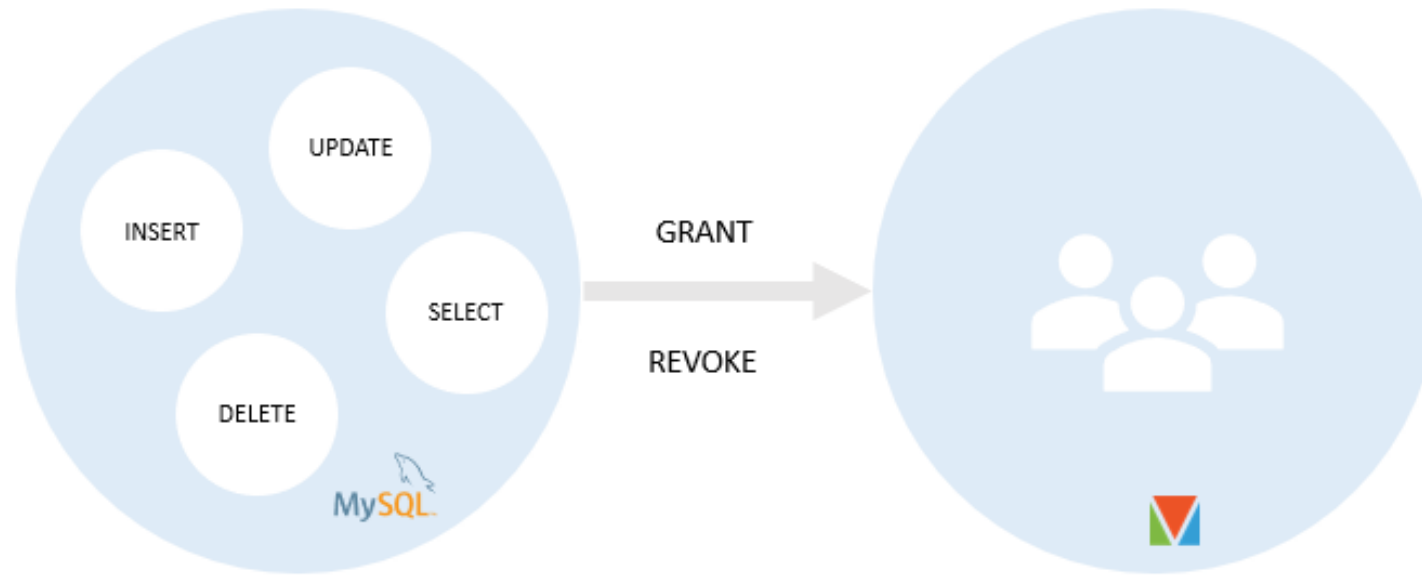
Privileges and Roles

- Privileges **granted** to a MySQL **account** determine **which operations the account can perform**.



Privileges and Roles

- Roles are served as a container or **collection of privileges**.



GRANT statement

Syntax

```
GRANT privilege [,privilege],..  
ON privilege_level  
TO account_name;
```

Example

```
mysql> GRANT SELECT  
      -> ON *.*  
      -> TO byte1@localhost;
```

```
Query OK, 0 rows affected (0.00 sec)
```

Privileges Provided by MySQL

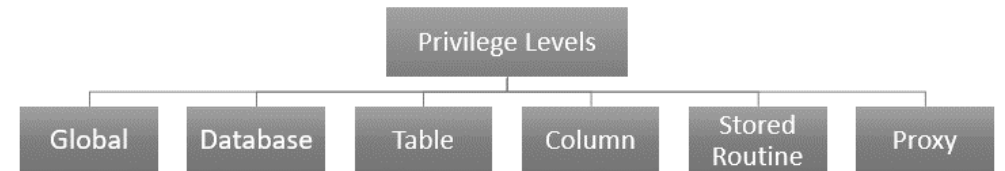
See details:

<https://dev.mysql.com/doc/refman/8.4/en/privileges-provided.html>

MySQL privilege levels

See details:

<https://www.mysqltutorial.org/mysql-administration/mysql-grant/>



Show GRANTS

Syntax

```
SHOW GRANTS FOR account_name;
```

Example

```
mysql> SHOW GRANTS FOR byte1@localhost;
```

```
+-----+
| Grants for byte1@localhost |
+-----+
| GRANT SELECT ON *.* TO `byte1`@`localhost` |
+-----+
1 row in set (0.00 sec)
```

CREATE ROLE statement

Syntax

```
CREATE ROLE  
role1,  
role2,  
role3;
```

Example

```
mysql> CREATE ROLE  
-> db_dev,  
-> db_read,  
-> db_write;
```

```
Query OK, 0 rows affected (0.00 sec)
```

Granting privileges to roles

Syntax (Same as Granting to user)

```
GRANT privilege [,privilege],..  
ON privilege_level  
TO role_name;
```

Example

```
mysql> GRANT INSERT, UPDATE, DELETE  
      -> ON *.*  
      -> TO db_write;  
Query OK, 0 rows affected (0.01 sec)
```

Assigning roles to user accounts

Syntax

```
GRANT role_name  
TO account_name;
```

Example

```
mysql> CREATE USER 'pj1_write'@localhost IDENTIFIED BY 'pj1_write';  
mysql> GRANT db_write  
      -> TO 'pj1_write'@localhost;  
mysql> SHOW GRANTS FOR 'pj1_write'@localhost;  
+-----+  
| Grants for pj1_write@localhost |  
+-----+  
| GRANT USAGE ON *.* TO `pj1_write`@`localhost` |  
| GRANT `db_write`@`%` TO `pj1_write`@`localhost` |  
+-----+  
2 rows in set (0.00 sec)
```

DROP statement

Syntax

```
DROP USER [IF EXISTS] account_name;
```

Ex.Check

```
mysql> SELECT user FROM mysql.user;
+-----+
| user                |
+-----+
| db_dev              |
| db_read             |
| db_write            |
| byte1               |
| mysql.infoschema    |
| mysql.session       |
| mysql.sys           |
| pj1_write          |
| root                |
+-----+
9 rows in set (0.00 sec)
```

Ex.Remove

```
mysql> DROP USER IF EXISTS
'pj1_write'@'localhost';

Query OK, 0 rows affected
(0.00 sec)
```

Ex.Re-Check

```
mysql> SELECT user FROM mysql.user;
+-----+
| user                |
+-----+
| db_dev              |
| db_read             |
| db_write            |
| byte1               |
| mysql.infoschema    |
| mysql.session       |
| mysql.sys           |
| root                |
+-----+
9 rows in set
(0.00 sec)
```

REVOKE statement

Syntax

```
REVOKE privilege [,privilege],..  
ON privilege_level  
FROM account_name;
```

Ex.Check

```
mysql> SHOW GRANTS FOR byte1@localhost;
```

```
+-----+  
| Grants for byte1@localhost |  
+-----+  
| GRANT SELECT ON *.* TO `byte1`@`localhost` |  
+-----+  
1 row in set (0.00 sec)
```

Ex.Re-Check

```
mysql> SHOW GRANTS FOR byte1@localhost;
```

```
+-----+  
| Grants for byte1@localhost |  
+-----+  
| GRANT USAGE ON *.* TO `byte1`@`localhost` |  
+-----+  
1 row in set (0.00 sec)
```

Ex.Remove

```
mysql> REVOKE SELECT ON *.* FROM byte1@localhost;  
Query OK, 0 rows affected (0.01 sec)
```

Note. The USAGE means that the user can log in to the database but has no privilege.

Using MySQL Workbench for User Account Management

Create a Database User Account

Go to “Administration” tab, then “Users and Privileges”

1

Administration Schemas

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges** **2**
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Local Instance 3306
Users and Privileges **3**

User Accounts

Select an account to edit or click [Add Account] to create a new one

User	From Host
itcs393	localhost
root	localhost

Login Account Limits Administrative Roles Schema Privileges

Login Name: You may create multiple accounts with the same name to connect from different hosts.

Authentication Type: For the standard password and/or host based authentication, select 'Standard'.

Limit to Hosts Matching: % and _ wildcards may be used

Password: Type a password to reset it.

Consider using a password with 8 or more characters with mixed case letters, numbers and punctuation marks.

Confirm Password: Enter password again to confirm.

Authentication String: Authentication plugin specific parameters.

Create a Database User Account

Login

Account Limits

Administrative Roles

Schema Privileges

Login Name:

itcs393

You may create multiple accounts with the same name to connect from different hosts.

Authentication Type:

caching_sha2_password

For the standard password and/or host based authentication, select 'Standard'.

Limit to Hosts Matching:

localhost

2 Change host to localhost

Password:

.....

Type a password to reset it.

Confirm Password:

.....

Enter password again to confirm.

Authentication String:

\$A\$005\$xfNQ0i#-}{6y1Z}ZFQFHbiLT

Authentication plugin specific parameters.

See the plugin documentation for valid values and details.

1 Define username

3 Put the password and confirm

Create a Database User Account

1

Select the Role *if needed*

Login | Account Limits | **Administrative Roles** | Schema Privileges

	Role	Description
☐	DBA	grants the rights to perform all tasks
☐	MaintenanceAdmin	grants rights needed to maintain server
☐	ProcessAdmin	rights needed to assess, monitor, and kill any u...
☐	UserAdmin	grants rights to create users logins and reset p...
☐	SecurityAdmin	rights to manage logins and grant and revoke s...
☐	MonitorAdmin	minimum set of rights needed to monitor server
☐	DBManager	grants full rights on all databases
☒	DBDesigner	rights to create and reverse engineer any data...
☐	ReplicationAdmin	rights needed to setup and manage replication
☐	BackupAdmin	minimal rights needed to backup any database

Global Privileges

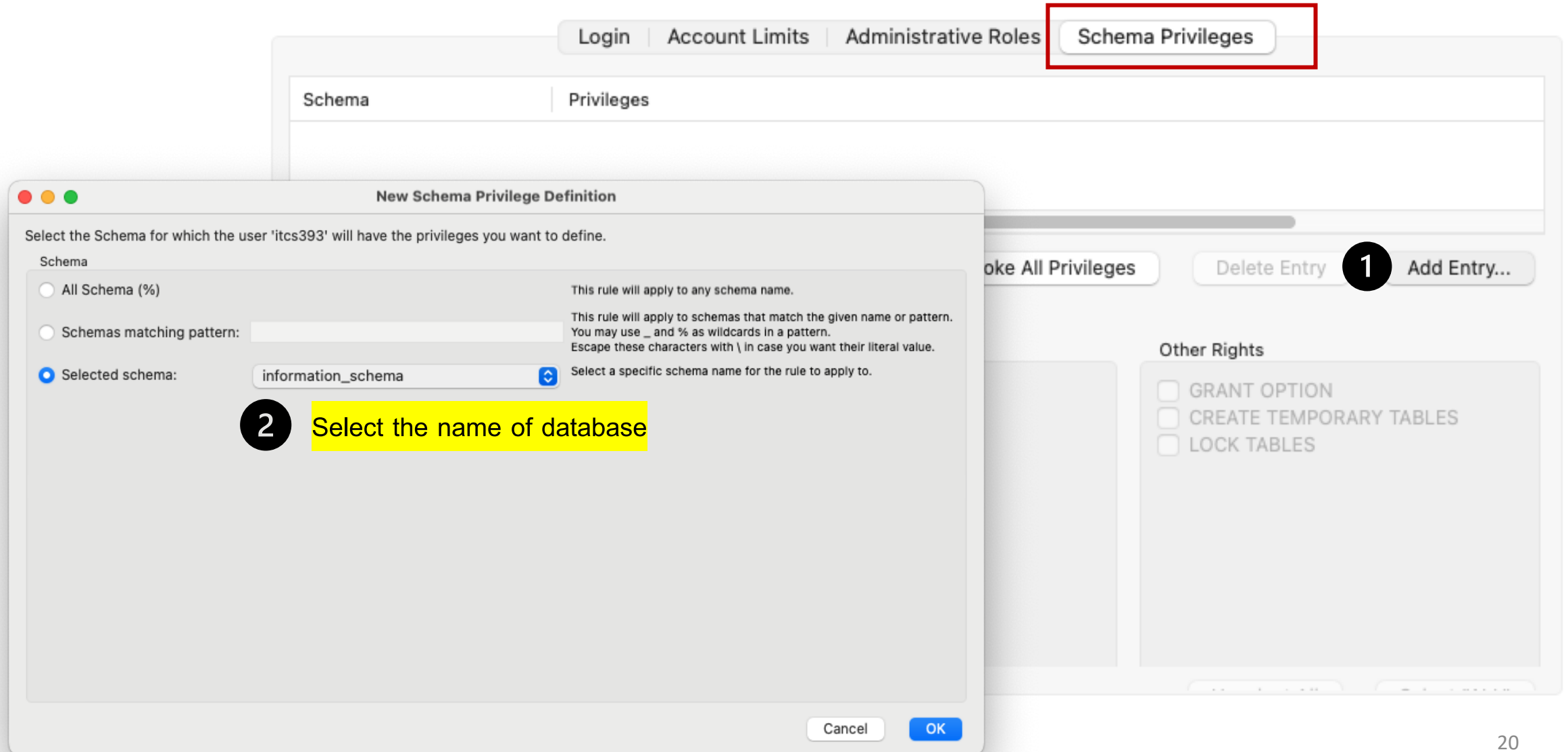
- ☒ ALTER
- ☒ ALTER ROUTINE
- ☒ CREATE
- ☒ CREATE ROUTINE
- ☐ CREATE TABLESPACE
- ☐ CREATE TEMPORARY TA
- ☐ CREATE USER
- ☒ CREATE VIEW
- ☐ DELETE
- ☐ DROP
- ☐ EVENT
- ☐ EXECUTE
- ☐ FILE

Revoke All Privileges

2

Select additional
privilege *if needed*

Create a Database User Account



Create a Database User Account

Login | Account Limits | Administrative Roles | **Schema Privileges**

Schema	Privileges
tinycollege	DELETE, INSERT, SELECT, UPDATE

Schema and Host fields may use % and _ wildcards.
The server will match specific entries before wildcarded ones.

Revoke All Privileges | Delete Entry | Add Entry...

The user 'itds241'@'localhost' will have the following access rights to the schema 'tinycollege':

Object Rights

- ☒ SELECT
- ☒ INSERT
- ☒ UPDATE
- ☒ DELETE
- ☐ EXECUTE
- ☐ SHOW VIEW

DDL Rights

- ☐ CREATE
- ☐ ALTER
- ☐ REFERENCES
- ☐ INDEX
- ☐ CREATE VIEW
- ☐ CREATE ROUTINE
- ☐ ALTER ROUTINE
- ☐ EVENT
- ☐ DROP
- ☐ TRIGGER

Other Rights

- ☐ GRANT OPTION
- ☐ CREATE TEMPORARY TABLES
- ☐ LOCK TABLES

Unselect All | Select "ALL"

3 Select Privileges

Create a Database User Account

Local instance 3306
Users and Privileges

User Accounts

User	From Host
itcs393	localhost
root	localhost

Details for account itcs393@localhost

Login | Account Limits | Administrative Roles | Schema Privileges

Schema	Privileges
tinycollege	DELETE, INSERT, SELECT, UPDATE

Schema and Host fields may use % and _ wildcards.
The server will match specific entries before wildcarded ones.

Revoke All Privileges | Delete Entry | Add Entry...

The user 'itcs393'@'localhost' will have the following access rights to the schema 'tinycollege':

Object Rights

- ☒ SELECT
- ☒ INSERT
- ☒ UPDATE
- ☒ DELETE
- ☐ EXECUTE
- ☐ SHOW VIEW

DDL Rights

- ☐ CREATE
- ☐ ALTER
- ☐ REFERENCES
- ☐ INDEX
- ☐ CREATE VIEW
- ☐ CREATE ROUTINE
- ☐ ALTER ROUTINE
- ☐ EVENT
- ☐ DROP
- ☐ TRIGGER

Other Rights

- ☐ GRANT OPTION
- ☐ CREATE TEMPORARY TABLES
- ☐ LOCK TABLES

Unselect All | Select "ALL"

Add Account | Delete | Refresh

Click "Apply"

Revert | Apply

Query on Workbench

